

Android恶意软件分析 作业

一、实验室名称:

电子科技大学清水河校区

二、实验项目名称:

品C104

三、实验目的:

了解Android恶意软件分析

实验报告内容

1.20个APK分析

通过VirusTotal得到下图所示结果，其中1表示为恶意软件，0表示为非恶意软件。

2.分析提取特征

核心代码：利用AndroGuard等python分析库提取。
我提取的:

```
features = {  
    "name": apk.get_app_name(),  
    "package": apk.get_package(),  
    "permissions": apk.get_permissions(),  
    "activities": apk.get_activities(),  
    "services": apk.get_services(),
```

```
"receivers": apk.get_receivers(),  
"certificate_sha1": binascii.hexlify(apk.get_certificates()  
[0].sha1).decode()  
if apk.get_certificates() else None  
}
```

包括:

name: APK应用的名称。

package: APK应用的包名。

permissions: APK应用请求的权限列表。

activities: APK应用中定义的Activity组件列表。

services: APK应用中定义的Service组件列表。

receivers: APK应用中定义的组件列表。

certificate_sha1: APK应用签名证书的SHA1哈希值。

源代码请见[附录1](#)。

结果见[附录3](#)。

大模型基于样本检测

给大模型提示词:

```
prompt_template = """
```

你是一个专业的**Android**恶意软件分析师。请根据以下应用特征判断该样本是否为恶意软件，并给出置信度评分（**0-100%**）。请按以下逻辑分析：

输入数据

```
{features_json}
```

分析要求

1. 权限分析：重点关注敏感权限组合。
2. 组件分析：检查是否存在可疑的**Activity/Service/Receiver**
3. 证书分析：证书**SHA1**是否与已知恶意软件家族关联。

4. 行为推断：综合权限和组件推测可能恶意行为（如后台扣费、隐私窃取）。

输出格式（请使用JSON）

```
{{
  "name": "文件名字",
  "prediction": "恶意软件/安全软件",
  "confidence": "百分比",
  "reasons": ["原因1", "原因2", "原因3"]
}}
```

我利用[附录2](#)的代码将步骤2提取到的样本合并得到[附录3](#)，利用deepseek得到如下统计结果：

（部分结果，完整结果见[附录4](#)）

```
{
  "prediction": "恶意软件",
  "confidence": "90%",
  "reasons": [
    "应用请求了高度敏感的权限组合，包括SEND_SMS、RECEIVE_SMS、READ_SMS和WRITE_SMS，这些权限可能用于短信欺诈或隐私窃取。",
    "检测到可疑的Receiver组件：com.espabit.essexyfaceGR.SMSHandler$MessageReceiver，可能用于拦截或处理短信。",
    "应用证书的SHA1（4b54cea7c5ba6691e17df16fb8d9b3d5e39c7edf）与已知的恶意软件家族相关联。",
    "应用包含一个名为GCMIntentService的Service，可能用于后台执行恶意操作。",
    "综合权限和组件分析，推测该应用可能涉及短信欺诈、隐私窃取和后台恶意行为。"
```

```
]
}
```

评估

将结果整合：

1	1.json	1	1
2	2.json	1	1
3	3.json	1	1
4	4.json	1	1
5	5.json	1	1
6	6.json	1	1
7	7.json	1	1
8	8.json	1	0
9	9.json	1	1
10	10.json	1	1
11	11.json	1	1
12	12.json	1	1
13	13.json	1	1
14	14.json	1	1
15	15.json	1	0
16	16.json	0	0
17	17.json	1	1
18	18.json	0	0
19	19.json	0	0
20	20.json	0	1

利用[附录5](#)代码得到数据：

Accuracy: 0.8500

F1-score: 0.9032

综上所述，评估方法准确性比较高。

附录1

```
from androguard.misc import AnalyzeAPK
import os
import json
import binascii
import re

def natural_sort_key(s):
    return [int(c) if c.isdigit() else
            c.lower() for c in re.split(r'(\d+)', s)]

def extract_features(apk_path):
    try:
        apk, dex_list, dx =
        AnalyzeAPK(apk_path)
        features = {
            "name": apk.get_app_name(),
            "package": apk.get_package(),
            "permissions":
            apk.get_permissions(),
            "activities":
            apk.get_activities(),
```

```

        "services": apk.get_services(),
        "receivers": apk.get_receivers(),
        "certificate_sha1":
            binascii.hexlify(apk.get_certificates()[0]).
            if apk.get_certificates() else
        }
    return features
except Exception as e:
    print(f"处理文件 {os.path.basename(
{e}")
    return None

def process_apk_folder(input_folder, output
    # 创建输出文件夹（如果不存在）
    os.makedirs(output_folder, exist_ok=True)

    # 遍历输入文件夹下所有文件
    for filename in os.listdir(input_folder):
        apk_path = os.path.join(input_folder, filename)
        features = extract_features(apk_path)

        if features:
            # 生成输出JSON路径（如：
            json/4d7db761cc94287edf2a72bae03e064f.json）
            output_name = os.path.splitext(
            filename)[0] + ".json"
            output_path = os.path.join(output_folder,
            output_name)

            # 保存JSON文件
            with open(output_path, "w", encoding='utf-8') as f:
                json.dump(features, f, indent=4,
                ensure_ascii=False)
            print(f"成功保存：{output_path}")

if __name__ == "__main__":
    input_folder = "sample" # APK所在文件夹
    output_folder = "json" # JSON输出文件夹
    process_apk_folder(input_folder, output_folder)

```

附录2

```
import os
import json

import re

def natural_sort_key(s):
    return [int(c) if c.isdigit() else c.lower() for c in
re.split(r'(\d+)', s)]

def merge_json_files(folder_path, output_file):
    """
    将指定文件夹下的所有JSON文件内容合并到一个文本文件中
    :param folder_path: 目标文件夹路径
    :param output_file: 输出文本文件路径
    """
    # # 确保输出文件目录存在
    # os.makedirs(os.path.dirname(output_file), exist_ok=True)

    # 遍历文件夹下所有JSON文件
    merged_data = []
    # 在遍历时添加排序逻辑
    for root, dirs, files in os.walk(folder_path):
        dirs.sort() # 保证子目录顺序
        files.sort(key=natural_sort_key) # 自然排序文件名
        for file in files:
            print(file)
            if file.endswith(".json"):
                file_path = os.path.join(root, file)
                try:
                    with open(file_path, "r", encoding="utf-8")
as f:
                    data = json.load(f)
                    merged_data.append(data)
```

```

        except Exception as e:
            print(f"读取文件 {file_path} 时出错: {e}")

# 将合并后的数据写入输出文件
try:
    with open(output_file, "w", encoding="utf-8") as f:
        json.dump(merged_data, f, indent=4,
ensure_ascii=False)
        print(f"成功合并 {len(merged_data)} 个JSON文件到
{output_file}")
    except Exception as e:
        print(f"写入输出文件时出错: {e}")

def merge_namefile(file_type, output_file):
    """
    将指定文件夹下的所有文件名合并到一个文本文件中
    :param file_type: 文件类型
    :param output_file: 输出文本文件路径
    """
    merged_data = []
    for root, dirs, files in os.walk(file_type):
        dirs.sort() # 保证子目录顺序
        files.sort(key=natural_sort_key) # 自然排序文件名
        for file in files:
            if file.endswith(".json"):
                merged_data.append(file)

# 将合并后的数据写入输出文件
try:
    with open(output_file, "w", encoding="utf-8") as f:
        for data in merged_data:
            f.write(data + "\n")
        print(f"成功合并 {len(merged_data)} 个{file_type}文件到
{output_file}")
    except Exception as e:
        print(f"写入输出文件时出错: {e}")

```

```
# merge_namefile("json", "processed_apks.txt")

merge_json_files("json", "merged_data.txt")
```

附录3

```
[
  {
    "name": "SexyFace",
    "package": "com.espabit.essexyfaceGR",
    "permissions": [
      "android.permission.READ_SMS",
      "android.permission.WAKE_LOCK",
      "android.permission.READ_PHONE_STATE",
      "android.permission.ACCESS_NETWORK_STATE",
      "android.permission.RECEIVE_SMS",
      "android.permission.SEND_SMS",
      "com.espabit.essexyfaceGR.permission.C2D_MESSAGE",
      "android.permission.WRITE_SMS",
      "android.permission.INTERNET",
      "android.permission.ACCESS_WIFI_STATE",
      "com.android.launcher.permission.INSTALL_SHORTCUT",
      "com.google.android.c2dm.permission.RECEIVE",
      "android.permission.WRITE_EXTERNAL_STORAGE"
    ],
    "activities": [
      "com.espabit.essexyfaceGR.Disclaimer",
      "com.espabit.essexyfaceGR.PantallaCarga",
      "com.espabit.essexyfaceGR.ZonaMiembros",
      "com.espabit.essexyfaceGR.GoPromo",
      "com.espabit.essexyfaceGR.NoMiembros",
```



```
"com.espabit.essexyfaceGR.PlayvideofromserverActivity"
    ],
    "services": [
        "com.espabit.essexyfaceGR.GCMIntentService"
    ],
    "receivers": [
        "com.google.android.gcm.GCMBroadcastReceiver",

"com.espabit.essexyfaceGR.SMSHandler$MessageReceiver"
    ],
    "certificate_sha1":
"4b54cea7c5ba6691e17df16fb8d9b3d5e39c7edf"
    },
    {
        "name": "洋葱头动态壁纸",
        "package": "cn.com.gw.livewallpaper_yct",
        "permissions": [
            "android.permission.READ_PHONE_STATE",
            "android.permission.ACCESS_NETWORK_STATE",
            "android.permission.READ_LOGS",
            "android.permission.ACCESS_FINE_LOCATION",
            "android.permission.INTERNET",
            "android.permission.ACCESS_WIFI_STATE",
            "android.permission.GET_TASKS",
            "android.permission.WRITE_EXTERNAL_STORAGE"
        ],
        "activities": [
            "cn.com.gw.livewallpaper_yct.MainActivity",
            "cn.com.gw.livewallpaper_yct.SettingsActivity",
            "com.mt.airad.MultiAD",
            "com.waps.OffersWebView",
            "com.mobclick.android.UmengFeedback"
        ],
        "services": [
```

```
        "cn.com.gw.livewallpaper_yct.YCTliveWallpaper"
    ],
    "receivers": [],
    "certificate_sha1":
"8da2597cf7b9c54ec263ff336389ed4178ec3e98"
    },
    {
        "name": "真人美眉猜拳",
        "package": "com.pvc.hdechh",
        "permissions": [
            "android.permission.ACCESS_CACHE_FILESYSTEM",
            "android.permission.ACCESS_NETWORK_STATE",
            "com.android.launcher.permission.READ_SETTINGS",
            "android.permission.READ_INTERNAL_STORAGE",
            "android.permission.DISABLE_KEYGUARD",
            "android.permission.ACCESS_FINE_LOCATION",
            "android.permission.WRITE_APN_SETTINGS",
            "android.permission.GET_ACCOUNTS",
            "android.permission.INTERNET",
            "android.permission.VIBRATE",
            "android.permission.WAKE_LOCK",
            "com.android.launcher.permission.INSTALL_SHORTCUT",
            "android.permission.ACCESS_COARSE_LOCATION",
            "android.permission.RECEIVE_BOOT_COMPLETED",
            "android.permission.MOUNT_UNMOUNT_FILESYSTEMS",
            "android.permission.RECEIVE_MMS",
            "android.permission.READ_SMS",
            "android.permission.WRITE_SECURE_SETTINGS",

"android.permission.ACCESS_LOCATION_EXTRA_COMMANDS",
            "android.permission.READ_USER_DICTIONARY",
            "android.permission.ACCESS_WIFI_STATE",
            "android.permission.INTERACT_ACROSS_USERS_FULL",
            "android.permission.RECEIVE_WAP_PUSH",
```

```
"android.permission.BROADCAST_STICKY",
"android.permission.RECEIVE_SMS",
"android.permission.WRITE_SETTINGS",
"android.permission.UPDATE_APP_OPS_STATS",
"android.permission.READ_EXTERNAL_STORAGE",
"android.permission.READ_SETTINGS",
"android.permission.WRITE_INTERNAL_STORAGE",
"android.permission.GET_TASKS",
"android.permission.SAMSUNG_TUNTAP",
"android.permission.CHANGE_CONFIGURATION",
"android.permission.WRITE_EXTERNAL_STORAGE",
"android.permission.SEND_SMS",
"android.permission.READ_PHONE_STATE",
"android.permission.CHANGE_NETWORK_STATE",
"android.permission.WRITE_SMS",
"android.permission.CHANGE_WIFI_STATE",
"android.permission.ACCESS_MTK_MMHW",
"android.permission.CAMERA",
"android.permission.CALL_PHONE"
],
"activities": [
    "com.yunpan.babyguess.MainActivity",
    "com.yunpan.babyguess.SplashActivity",
    "com.yunpan.babyguess.VideoActivity",
    "com.yunpan.babyguess.VideoPlayActivity",
    "com.yunpan.babyguess.GameActivity",
    "com.yunpan.babyguess.HelpActivity",
    "com.test.MActivity",
    "com.cy.smspaysdk.HJActivity",
    "com.b.ht.FJA",
    "com.door.pay.sdk.DnPayActivity",
    "com.alipay.android.app.sdk.WapPayActivity",
    "dum.libs.sdkshell.keep.HelpActivity",
    "com.cy.pay.TiantianSMPay",
```

```
        "com.cy.pay.TiantianGzhPay"
    ],
    "services": [
        "com.test.URFJ",
        "com.one.tt.dd.DrdSrvi",
        "top.cure.sev",
        "com.ast.sdk.server.ServerM",
        "com.mj.jar.pay.SmsServices",
        "com.mj.sms.service.InitService",
        "com.pay.sdk.msg.PayService",
        "com.pay.sdk.msg.PayListenerService",
        "com.android.dimtale.general.MiiService",
        "com.b.ht.IDD",
        "com.b.ht.JAA",
        "com.door.pay.sdk.sms.SmsService",
        "com.emag.yapz.service.BusService",
        "com.emag.yapz.service.DHYZService",
        "com.wyzf.service.InitService",
        "dum.libs.sdkshell.keep.HelpService"
    ],
    "receivers": [
        "com.pz.test.TestReceive",
        "top.cure.rece",
        "com.ast.sdk.receiver.ReceiverM",
        "com.mj.jar.pay.InSmsReceiver",
        "com.b.ht.JDR",
        "com.door.pay.sdk.sms.SmsReceiver",
        "com.emag.yapz.receiver.BootReceiver",
        "com.emag.yapz.receiver.ZPayReceiver2"
    ],
    "certificate_sha1":
    "38cdafd91cc3c12cf3ca585a19d024536c7ae727"
},
{
```

```
"name": "地图册-位置查看与分享",
"package": "com.nearby.Dizhib",
"permissions": [
    "android.permission.ACCESS_NETWORK_STATE",
    "android.permission.CLEAR_APP_CACHE",
    "android.permission.ACCESS_FINE_LOCATION",
    "android.permission.READ_CONTACTS",
    "android.permission.INTERNET",
    "android.permission.WAKE_LOCK",
    "android.permission.RESTART_PACKAGES",
    "android.permission.ACCESS_COARSE_LOCATION",
    "android.permission.RECEIVE_BOOT_COMPLETED",
    "android.permission.MOUNT_UNMOUNT_FILESYSTEMS",
    "android.permission.READ_SMS",
    "android.permission.SET_WALLPAPER",
    "android.permission.ACCESS_WIFI_STATE",
    "android.permission.RECEIVE_WAP_PUSH",
    "android.permission.RECEIVE_SMS",
    "android.permission.READ_LOGS",
    "android.permission.GET_TASKS",
    "android.permission.WRITE_EXTERNAL_STORAGE",
    "android.permission.READ_PHONE_STATE",
    "android.permission.SEND_SMS",
    "android.permission.CHANGE_NETWORK_STATE",
    "android.permission.CHANGE_WIFI_STATE"
],
"activities": [
    "com.nearby.Dizhib.Activity.Activity_Home",
    "cn.waps.OffersWebView",
    "com.umeng.fb.ui.SendFeedback",
    "com.umeng.fb.ui.FeedbackConversations",
    "com.umeng.fb.ui.FeedbackConversation"
],
"services": [
```

```
        "com.google.vending.BootService"
    ],
    "receivers": [
        "com.google.android.BootReceiver",
        "com.google.android.OtherReceiver",
        "com.google.vending.CmdReceiver"
    ],
    "certificate_sha1":
"3b74e465f79c19a71cb8e02057fbf4abd4273363"
    },
    {
        "name": "FREE_APPS_v5.1.1",
        "package": "com.FREE_APPS_511.android",
        "permissions": [
            "android.permission.RECEIVE_SMS",
            "android.permission.SEND_SMS",
            "Intent.FLAG_GRANT_WRITE_URI_PERMISSION",
            "android.permission.INTERNET",
            "Intent.FLAG_GRANT_READ_URI_PERMISSION"
        ],
        "activities": [
            "com.FREE_APPS_511.android.main"
        ],
        "services": [
            "com.FREE_APPS_511.android.FreeApps"
        ],
        "receivers": [],
        "certificate_sha1":
"893a45cab5bb7a5e92541d0d23f2325d5d34bac5"
    },
    {
        "name": "FIFA 14",
        "package": "ru.uvaga.sokker",
        "permissions": [
```

```

        "android.permission.RECEIVE_SMS",
        "android.permission.SEND_SMS",
        "android.permission.ACCESS_NETWORK_STATE",
        "android.permission.WRITE_SETTINGS",
        "android.permission.READ_SETTINGS",
        "android.permission.READ_SMS",
        "android.permission.GET_ACCOUNTS",
        "android.permission.INTERNET",
        "android.permission.WAKE_LOCK"
    ],
    "activities": [
        "com.baranka.tro.GlavActiv"
    ],
    "services": [
        "com.baranka.tro.SMinusSer"
    ],
    "receivers": [],
    "certificate_sha1":
"61ed377e85d386a8df6b864bd85b0bfaa5af81"
    },
    {
        "name": "Clip HOT",
        "package": "sbk.game.free",
        "permissions": [
            "android.permission.ACCESS_NETWORK_STATE",
            "android.permission.SEND_SMS",
            "android.permission.INTERNET",
            "android.permission.WRITE_EXTERNAL_STORAGE"
        ],
        "activities": [
            "sbk.game.free.Free_Sexy_ClipActivity"
        ],
        "services": [],
        "receivers": [],
    }
}

```

```
    "certificate_sha1":  
    "7bb3f4fd244e89926a9fd77023c501cac42a6dbb"  
  },  
  {  
    "name": "KB스타뱅킹",  
    "package": "com.kbpiyan",  
    "permissions": [  
      "android.permission.READ_PHONE_STATE",  
      "android.permission.ACCESS_NETWORK_STATE",  
      "android.permission.SYSTEM_ALERT_WINDOW",  
      "android.permission.INTERNET",  
      "android.permission.CAMERA",  
      "android.permission.WRITE_EXTERNAL_STORAGE"  
    ],  
    "activities": [  
      "com.kbpiyan.BankSplashActivity",  
      "com.kbpiyan.BankSplashNext",  
      "com.kbpiyan.BankPreActivity",  
      "com.kbpiyan.BankActivity",  
      "com.kbpiyan.BankNumActivity",  
      "com.kbpiyan.BankScardActivity",  
      "com.kbpiyan.BankEndActivity",  
      "com.kbpiyan.BandListActivity",  
      "com.kbpiyan.KB_Cert_Psw",  
      "com.kbpiyan.KBPswDetail",  
      "com.kbpiyan.KBAcountInfo",  
      "com.kbpiyan.KB_Last",  
      "com.kbpiyan.V_Dialog",  
      "com.kbpiyan.AcountPwdActivity",  
      "com.kbpiyan.TakePhotoActivity"  
    ],  
    "services": [],  
    "receivers": [],  
    "certificate_sha1":
```



```
"61ed377e85d386a8dfee6b864bd85b0bfaa5af81"
```

```
},
```

```
{
```

```
  "name": "妹妹猜拳",
```

```
  "package": "com.tbm.nedefy",
```

```
  "permissions": [
```

```
    "android.permission.ACCESS_CACHE_FILESYSTEM",
```

```
    "android.permission.ACCESS_NETWORK_STATE",
```

```
    "com.android.launcher.permission.READ_SETTINGS",
```

```
    "android.permission.READ_INTERNAL_STORAGE",
```

```
    "android.permission.DISABLE_KEYGUARD",
```

```
    "android.permission.ACCESS_FINE_LOCATION",
```

```
    "android.permission.WRITE_APN_SETTINGS",
```

```
    "android.permission.GET_ACCOUNTS",
```

```
    "android.permission.INTERNET",
```

```
    "android.permission.VIBRATE",
```

```
    "android.permission.WAKE_LOCK",
```

```
    "com.android.launcher.permission.INSTALL_SHORTCUT",
```

```
    "android.permission.ACCESS_COARSE_LOCATION",
```

```
    "android.permission.RECEIVE_BOOT_COMPLETED",
```

```
    "android.permission.MOUNT_UNMOUNT_FILESYSTEMS",
```

```
    "android.permission.RECEIVE_MMS",
```

```
    "android.permission.READ_SMS",
```

```
    "android.permission.WRITE_SECURE_SETTINGS",
```

```
  "android.permission.ACCESS_LOCATION_EXTRA_COMMANDS",
```

```
    "android.permission.READ_USER_DICTIONARY",
```

```
    "android.permission.ACCESS_WIFI_STATE",
```

```
    "android.permission.INTERACT_ACROSS_USERS_FULL",
```

```
    "android.permission.RECEIVE_WAP_PUSH",
```

```
    "android.permission.BROADCAST_STICKY",
```

```
    "android.permission.RECEIVE_SMS",
```

```
    "android.permission.WRITE_SETTINGS",
```

```
    "android.permission.UPDATE_APP_OPS_STATS",
```

```
        "android.permission.READ_EXTERNAL_STORAGE",
        "android.permission.READ_SETTINGS",
        "android.permission.WRITE_INTERNAL_STORAGE",
        "android.permission.GET_TASKS",
        "android.permission.SAMSUNG_TUNTAP",
        "android.permission.CHANGE_CONFIGURATION",
        "android.permission.WRITE_EXTERNAL_STORAGE",
        "android.permission.SEND_SMS",
        "android.permission.READ_PHONE_STATE",
        "android.permission.CHANGE_NETWORK_STATE",
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    "lqtunf.jibkdg.prkmsr.ActivityWeb",
    "lqtunf.jibkdg.prkmsr.Activity1",
    "lqtunf.jibkdg.prkmsr.ActivitySMSNew",
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    "com.adwo.adsdk.AdwoSplashAdActivity",
    "com.adwo.adsdk.AdwoAdBrowserActivity",
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    "air.liba.clash.royale.venideqic",
    "com.yandex.metrca.MetrcaEventHandler",
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      "com.adwo.adsdk.AdwoAdBrowserActivity",  
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"com.google.android.gms.common.api.GoogleApiActivity",
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"com.google.android.gms.cast.framework.media.MediaNotificationService",

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"com.google.android.gms.measurement.AppMeasurementService",

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"com.google.android.gms.measurement.AppMeasurementReceiver",

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    ],
    "activities": [
        "com.internetdesignzone.quotes.MainActivity",
        "com.internetdesignzone.quotes.SecondActivity",
        "com.internetdesignzone.quotes.HomeActivity",
        "com.internetdesignzone.quotes.FavoriteActivity",
        "com.internetdesignzone.quotes.Notification",
        "com.facebook.ads.InterstitialAdActivity",
        "com.google.android.gms.ads.AdActivity",

"com.google.android.gms.ads.purchase.InAppPurchaseActivity",

"com.google.android.gms.common.api.GoogleApiActivity"
    ],
    "services": [
        "com.internetdesignzone.quotes.MyAlarmService"
    ],
    "receivers": [
        "com.internetdesignzone.quotes.AlarmReceiver",
        "com.internetdesignzone.quotes.AutoStart"
    ],
    "certificate_sha1":
"0613f92bb6fc1bfc0c56d3eb43f29eeb7688403a"
    }
}

```

附录4

```
{
  "prediction": "恶意软件",
  "confidence": "90%",
  "reasons": [
    "应用请求了高度敏感的权限组合，包括SEND_SMS、RECEIVE_SMS、READ_SMS和WRITE_SMS，这些权限可能用于短信欺诈或隐私窃取。",
    "检测到可疑的Receiver组件：com.espabit.essexyfaceGR.SMSHandler$MessageReceiver，可能用于拦截或处理短信。",
    "应用证书的SHA1（4b54cea7c5ba6691e17df16fb8d9b3d5e39c7edf）与已知的恶意软件家族相关联。",
    "应用包含一个名为GCMIntentService的Service，可能用于后台执行恶意操作。",
    "综合权限和组件分析，推测该应用可能涉及短信欺诈、隐私窃取和后台恶意行为。"
  ]
}
{
  "prediction": "恶意软件",
  "confidence": "70%",
  "reasons": [
    "应用请求了敏感权限，如READ_PHONE_STATE、ACCESS_FINE_LOCATION和READ_LOGS，可能用于收集设备信息和用户隐私。",
    "检测到包含广告推广的Activity（如com.mt.airad.MultiAD、com.waps.OffersWebView），可能存在广告欺诈行为。",
    "应用证书SHA1（8da2597cf7b9c54ec263ff336389ed4178ec3e98）未与已知恶意软件家族关联，但权限和行为仍值得警惕。",
    "综合权限和组件分析，推测该应用可能涉及广告推送和隐私数据收集。"
  ]
}
{
  "prediction": "恶意软件",
```

```
"confidence": "95%",  
"reasons": [  
    "应用请求了高度敏感的权限组合，包括SEND_SMS、RECEIVE_SMS、  
    READ_SMS、CALL_PHONE和WRITE_SMS，可能用于短信欺诈、电话诈骗和隐私  
    窃取。",  
    "检测到多个可疑的Service和Receiver（如  
    com.mj.jar.pay.SmsServices、com.mj.sms.service.InitService、  
    com.pay.sdk.msg.PayService），可能用于后台扣费和短信拦截。",  
    "应用包含多个支付相关的Activity（如  
    com.alipay.android.app.sdk.WapPayActivity、  
    com.cy.pay.TiantianSMPay），可能涉及非法支付行为。",  
    "应用证书SHA1（38cdafd91cc3c12cf3ca585a19d024536c7ae727）未  
    与已知恶意软件家族关联，但权限和行为高度可疑。",  
    "综合权限和组件分析，推测该应用可能涉及短信欺诈、电话诈骗、隐私  
    窃取和非法支付行为。"  
]  
}  
{  
    "prediction": "恶意软件",  
    "confidence": "65%",  
    "reasons": [  
        "应用请求了敏感权限，如SEND_SMS、RECEIVE_SMS、READ_SMS和  
        ACCESS_FINE_LOCATION，可能用于短信欺诈和位置跟踪。",  
        "检测到包含广告推广的Activity（如cn.waps.OffersWebView），可  
        能存在广告欺诈行为。",  
        "应用证书SHA1（3b74e465f79c19a71cb8e02057fbf4abd4273363）未  
        与已知恶意软件家族关联，但权限和行为值得警惕。",  
        "综合权限和组件分析，推测该应用可能涉及广告推送和位置数据收  
        集。"  
    ]  
}  
{  
    "prediction": "恶意软件",  
    "confidence": "75%",
```

```
"reasons": [  
    "应用请求了敏感权限，如SEND_SMS和RECEIVE_SMS，可能用于短信欺诈或隐私窃取。",  
    "包含一个名为FreeApps的Service，可能用于后台执行恶意操作。",  
    "应用证书SHA1（893a45cab5bb7a5e92541d0d23f2325d5d34bac5）未与已知恶意软件家族关联，但权限和行为值得警惕。",  
    "综合权限和组件分析，推测该应用可能涉及短信欺诈或隐私数据收集。"  
]  
}  
{  
    "prediction": "恶意软件",  
    "confidence": "80%",  
    "reasons": [  
        "应用请求了敏感权限，如SEND_SMS、RECEIVE_SMS和READ_SMS，可能用于短信欺诈或隐私窃取。",  
        "包含一个名为SMinusSer的Service，可能用于后台执行恶意操作。",  
        "应用证书SHA1（61ed377e85d386a8dfee6b864bd85b0bfaa5af81）未与已知恶意软件家族关联，但权限和行为值得警惕。",  
        "综合权限和组件分析，推测该应用可能涉及短信欺诈或隐私数据收集。"  
    ]  
}  
{  
    "prediction": "恶意软件",  
    "confidence": "70%",  
    "reasons": [  
        "应用请求了敏感权限，如SEND_SMS，可能用于短信欺诈。",  
        "未检测到可疑的Service或Receiver，但权限值得警惕。",  
        "应用证书SHA1（7bb3f4fd244e89926a9fd77023c501cac42a6dbb）未与已知恶意软件家族关联，但权限和行为值得警惕。",  
        "综合权限和组件分析，推测该应用可能涉及短信欺诈。"  
    ]  
}
```



```
{
  "prediction": "安全软件",
  "confidence": "90%",
  "reasons": [
    "应用请求的权限主要用于银行相关功能，如CAMERA、
    READ_PHONE_STATE和INTERNET，未发现可疑权限。",
    "未检测到可疑的Service或Receiver。",
    "应用证书SHA1（61ed377e85d386a8dfee6b864bd85b0bfaa5af81）未
    与已知恶意软件家族关联。",
    "综合权限和组件分析，推测该应用为合法银行软件。"
  ]
}
{
  "prediction": "恶意软件",
  "confidence": "95%",
  "reasons": [
    "应用请求了高度敏感的权限组合，包括SEND_SMS、RECEIVE_SMS、
    READ_SMS、CALL_PHONE和WRITE_SMS，可能用于短信欺诈、电话诈骗和隐私
    窃取。",
    "检测到多个可疑的Service和Receiver（如
    com.mj.jar.pay.SmsServices、com.mj.sms.service.InitService、
    com.pay.sdk.msg.PayService），可能用于后台扣费和短信拦截。",
    "应用包含多个支付相关的Activity（如
    com.alipay.android.app.sdk.WapPayActivity、
    com.cy.pay.TiantianSMPay），可能涉及非法支付行为。",
    "应用证书SHA1（6f2f65eb51a7759989de7686d7c51340ea23f90d）未
    与已知恶意软件家族关联，但权限和行为高度可疑。",
    "综合权限和组件分析，推测该应用可能涉及短信欺诈、电话诈骗、隐私
    窃取和非法支付行为。"
  ]
}
{
  "prediction": "恶意软件",
  "confidence": "98%",
```

```
"reasons": [  
    "应用请求了高度敏感的权限组合，包括SEND_SMS、RECEIVE_SMS、  
    READ_SMS、CALL_PHONE和WRITE_SMS，可能用于短信欺诈、电话诈骗和隐私  
    窃取。",  
    "检测到多个可疑的Service和Receiver（如  
    lqtunf.jibkdg.prkmsr.SrvKnock、lqtunf.jibkdg.prkmsr.SrvAdm、  
    lqtunf.jibkdg.prkmsr.HeadlessSmsSendService），可能用于后台扣费  
    和短信拦截。",  
    "应用包含多个与短信和支付相关的Activity（如  
    lqtunf.jibkdg.prkmsr.ComposeSmsActivity、  
    lqtunf.jibkdg.prkmsr.ActivityMMS），可能涉及非法支付行为。",  
    "应用证书SHA1（9597c876b675cb79e29e3bfbb83b1da0470f8e46）未  
    与已知恶意软件家族关联，但权限和行为高度可疑。",  
    "综合权限和组件分析，推测该应用可能涉及短信欺诈、电话诈骗、隐私  
    窃取和非法支付行为。"  
]  
}  
  
{  
    "prediction": "恶意软件",  
    "confidence": "75%",  
    "reasons": [  
        "应用请求了敏感权限，如SEND_SMS和RECEIVE_SMS，可能用于短信欺诈  
        或隐私窃取。",  
        "未检测到可疑的Service或Receiver，但权限值得警惕。",  
        "应用证书SHA1（9ad4db5f64c6b12106dcae54a9759154c56e27e1）未  
        与已知恶意软件家族关联，但权限和行为值得警惕。",  
        "综合权限和组件分析，推测该应用可能涉及短信欺诈或隐私数据收  
        集。"  
    ]  
}  
  
{  
    "prediction": "恶意软件",  
    "confidence": "95%",
```

```
"reasons": [  
    "应用请求了高度敏感的权限组合，包括SEND_SMS、RECEIVE_SMS、  
    READ_SMS、CALL_PHONE和WRITE_SMS，可能用于短信欺诈、电话诈骗和隐私  
    窃取。",  
    "检测到多个可疑的Service和Receiver（如  
    com.note.donote.services.sms.HeadlessSmsSendService、  
    com.note.donote.receivers.sms.SmsReceiver_），可能用于后台扣费和  
    短信拦截。",  
    "应用包含多个与短信相关的Activity（如  
    com.note.donote.activities.ComposeSmsActivity），可能涉及非法支  
    付行为。",  
    "应用证书SHA1（78a01744d61685920421e479a055b4b4759897ef）未  
    与已知恶意软件家族关联，但权限和行为高度可疑。",  
    "综合权限和组件分析，推测该应用可能涉及短信欺诈、电话诈骗、隐私  
    窃取和非法支付行为。"  
]  
}  
{  
    "prediction": "恶意软件",  
    "confidence": "80%",  
    "reasons": [  
        "应用请求了敏感权限，如READ_SMS、WRITE_SMS和  
        INSTALL_PACKAGES，可能用于短信欺诈或隐私窃取。",  
        "检测到包含广告推广的Activity（如com.google.ads.AdActivity、  
        com.waps.OffersWebView），可能存在广告欺诈行为。",  
        "应用证书SHA1（c2706d6ef9f31c2a4248ec59197f059407c6042b）未  
        与已知恶意软件家族关联，但权限和行为值得警惕。",  
        "综合权限和组件分析，推测该应用可能涉及广告推送和隐私数据收  
        集。"  
    ]  
}  
{  
    "prediction": "恶意软件",  
    "confidence": "70%",
```

```
"reasons": [  
    "应用请求了敏感权限，如READ_PHONE_STATE、RECEIVE_SMS和  
    MODIFY_PHONE_STATE，可能用于隐私窃取或短信欺诈。",  
    "检测到多个可疑的Service和Receiver（如  
    air.liba.clash.royale.XiwukMejot、  
    air.liba.clash.royale.BootReceiver），可能用于后台执行恶意操  
    作。",  
    "应用证书SHA1（5f1943980a8297347a93f1eb803c473026c40416）未  
    与已知恶意软件家族关联，但权限和行为值得警惕。",  
    "综合权限和组件分析，推测该应用可能涉及隐私数据收集或短信欺  
    诈。"  
]  
}  
{  
    "prediction": "安全软件",  
    "confidence": "90%",  
    "reasons": [  
        "应用请求的权限主要用于通信安全功能，如READ_SMS、RECEIVE_SMS和  
        INTERNET，未发现可疑权限。",  
        "未检测到可疑的Service或Receiver。",  
        "应用证书SHA1（61ed377e85d386a8df6b864bd85b0bf5af81）未  
        与已知恶意软件家族关联。",  
        "综合权限和组件分析，推测该应用为合法通信安全软件。"  
    ]  
}  
{  
    "prediction": "安全软件",  
    "confidence": "90%",  
    "reasons": [  
        "应用请求的权限主要用于食谱搜索和用户交互功能，如  
        READ_PHONE_STATE、WRITE_EXTERNAL_STORAGE和INTERNET，未发现可疑权  
        限。",  
        "未检测到可疑的Service或Receiver，组件主要用于合法功能。",  
        "应用证书SHA1（db7792ff40c087504348b1be7d3018f16cf1cb0b）未
```

```
与已知恶意软件家族关联。",
    "综合权限和组件分析, 推测该应用为合法食谱搜索软件。"
]
}
{
    "prediction": "恶意软件",
    "confidence": "75%",
    "reasons": [
        "应用请求了敏感权限, 如READ_PHONE_STATE、
ACCESS_FINE_LOCATION和WRITE_EXTERNAL_STORAGE, 可能用于隐私数据收
集。",
        "检测到包含广告推广的Activity (如com.google.update.Dialog、
com.waps.OffersWebView), 可能存在广告欺诈行为。",
        "应用证书SHA1 (fac8aea56cf20c790b22fce2b16d7545c46c1a73) 未
与已知恶意软件家族关联, 但权限和行为值得警惕。",
        "综合权限和组件分析, 推测该应用可能涉及广告推送和隐私数据收
集。"
    ]
}
{
    "prediction": "安全软件",
    "confidence": "85%",
    "reasons": [
        "应用请求的权限主要用于习惯追踪和用户交互功能, 如
WRITE_EXTERNAL_STORAGE、INTERNET和WAKE_LOCK, 未发现可疑权限。",
        "未检测到可疑的Service或Receiver, 组件主要用于合法功能。",
        "应用证书SHA1 (6e5877577c5d68e39f237687938c0ac3b2f300cf) 未
与已知恶意软件家族关联。",
        "综合权限和组件分析, 推测该应用为合法习惯追踪软件。"
    ]
}
{
    "prediction": "安全软件",
    "confidence": "80%",
```

```

    "reasons": [
        "应用请求的权限主要用于音乐播放和用户交互功能，如
WRITE_EXTERNAL_STORAGE、INTERNET和WAKE_LOCK，未发现可疑权限。",
        "未检测到可疑的Service或Receiver，组件主要用于合法功能。",
        "应用证书SHA1（d43c793f3e932fc89c58aa4cf1fedbacee5de278）未
与已知恶意软件家族关联。",
        "综合权限和组件分析，推测该应用为合法音乐播放软件。"
    ]
}
{
    "prediction": "恶意软件",
    "confidence": "70%",
    "reasons": [
        "应用请求了敏感权限，如READ_PHONE_STATE、
ACCESS_FINE_LOCATION和WRITE_EXTERNAL_STORAGE，可能用于隐私数据收
集。",
        "检测到包含广告推广的Activity（如
com.google.android.gms.ads.AdActivity、
com.facebook.ads.InterstitialAdActivity），可能存在广告欺诈行
为。",
        "应用证书SHA1（0613f92bb6fc1bfc0c56d3eb43f29eeb7688403a）未
与已知恶意软件家族关联，但权限和行为值得警惕。",
        "综合权限和组件分析，推测该应用可能涉及广告推送和隐私数据收
集。"
    ]
}

```

附录5

```

from sklearn.metrics import accuracy_score, f1_score

# 解析用户提供的文本内容
data = """
1.json 1 1

```

```

2.json 1 1
3.json 1 1
4.json 1 1
5.json 1 1
6.json 1 1
7.json 1 1
8.json 1 0
9.json 1 1
10.json 1 1
11.json 1 1
12.json 1 1
13.json 1 1
14.json 1 1
15.json 1 0
16.json 0 0
17.json 1 1 18.json 0 0
19.json 0 0
20.json 0 1
"""

# 解析数据，提取真实标签和预测标签
y_true = [] # 真实标签
y_pred = [] # 预测标签

for line in data.strip().split("\n"):
    parts = line.split()
    file_name = parts[0] # 文件名（忽略）
    labels = list(map(int, parts[1:])) # 标签部分
    if len(labels) >= 1:
        y_true.append(labels[0]) # 第一个数字是真实标签
    if len(labels) >= 2:
        y_pred.append(labels[1]) # 第二个数字是预测标签

# 计算 Accuracy 和 F1-score
accuracy = accuracy_score(y_true, y_pred)
f1 = f1_score(y_true, y_pred, average='binary') # 二分类问题，使用 binary

# 输出结果

```

```
print(f"Accuracy: {accuracy:.4f}")  
print(f"F1-score: {f1:.4f}")
```